



Buildings.city: Scalable urban building energy modeling and carbon emissions mapping using open archetype templates

Pengdi Lyu, Hongyao Wong, Royden Soh, Tao Wang, Yu Qian Ang^{*}

College of Design & Engineering, National University of Singapore, Singapore

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ABSTRACT

Cities and municipalities often struggle to scale building-level energy and carbon assessments to the urban scale, due to incomplete data, inconsistent input quality, and a lack of benchmarks. This study presents Buildings.city, an open-source Urban Building Energy Modeling (UBEM) framework and reproducible toolkit for city-scale carbon emissions accounting and mapping. The platform integrates globally available multi-source geospatial data, machine-learning-based archetype inference, and archetype-specific energy simulations (along with the associated templates) to generate detailed building-level carbon maps. To demonstrate the framework's adaptability to diverse urban contexts, the GitHub repository provides a baseline example of Zurich, and the entire toolkit was validated through a city-scale proof of concept in Singapore (Buildings.sg). In the full deployment, a predictive model achieved >75% accuracy in inferring missing building archetypes, addressing data gaps in OpenStreetMap (OSM). We also established a set of open-source building energy modeling packages and simulation templates for 23 archetypes, along with a web platform that visualizes simulated carbon emissions across approximately 120,000 buildings to support interactive analysis and policy decision-making. Ultimately, through a modular architecture that integrates diverse datasets, the Buildings.city toolkit provides a transparent, adaptable baseline for universal application.

1. Introduction

Cities have become focal points of climate risk. By 2050, a 2 °C increase in global temperature is projected to expose approximately 2.7 billion people to moderate or high climate risks, with the vast majority (viz., 91% to 98%) living in Africa and Asia ([United Nations Human Settlements Programme, 2024](#)). Within this context, the built environment is a critical driver of emissions. The building and construction sector accounts for approximately 37% of global energy-related carbon emissions, with around 27% from operational energy and nearly 10% from embodied carbon ([UNEP, 2024](#)). As the global urban population is projected to double by 2050, metropolitan areas are expected to account for over 70% of total emissions ([Ang, Berzolla, Letellier-Duchesne, et al., 2023](#)). The International Energy Agency ([IEA, 2024](#)) identifies the

building sector as a key area for decarbonization. However, energy intensity in this domain must fall by 37% from 2015 levels by 2030 ([UNFCCC, 2015](#)). As of 2022, the reduction achieved was only 3.5%, leaving the sector nearly 15 percentage points behind the necessary trajectory ([UNEP, 2024](#)). This shortfall highlights the need to accelerate climate action in the built environment.

While governments, policymakers, and planners face mounting pressure to decarbonize the built environment, existing efforts are often constrained by the fragmented nature of urban data, inconsistent modeling approaches, and the limited scalability of conventional building performance simulation tools ([Frayssinet et al., 2018](#); [Nidam, Irani, Bemis, & Reinhart, 2023](#); [Reinhart & Davila, 2016](#)). Although individual buildings can be assessed using high-fidelity, physics-based simulations, applying such methods across entire cities remains a

Abbreviations: ECF, Emission Conversion Factor; EPD, Environmental Product Declaration; EPW, EnergyPlus Weather File; EUI, Energy Use Intensity; GIS, Geographic Information System; GMM, Gaussian Mixture Model; GFA, Gross Floor Area; HDB, Housing and Development Board; HVAC, Heating, Ventilation, and Air Conditioning; IDF, EnergyPlus Input Data File; IEA, International Energy Agency; IHL, Institute of Higher Learning; IQR, Interquartile Range; LCA, Life Cycle Assessment; LPD, Lighting Power Density; NMBE, Normalized Mean Bias Error; OSM, OpenStreetMap; PAMC, Prefabrication And Modular Construction; PDF, Probability Density Function; SMOTE, Synthetic Minority Over-sampling Technique; TMY, Typical Meteorological Year; UBEM, Urban Building Energy Modeling.

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^{*} Corresponding author.

E-mail address: yuqian@nus.edu.sg (Y.Q. Ang).

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challenge (Hong, Chen, Luo, Luo, & Lee, 2020). For example, to simulate a single building's energy use, one typically needs detailed inputs such as window-to-wall ratios, HVAC system types, envelope characteristics, occupancy patterns translated into hourly schedules, and lighting and equipment power densities. The need for scalable, transparent, and locally relevant datasets and tools is particularly urgent.

Against this backdrop, the growing availability of open geospatial datasets – such as OpenStreetMap, government registries, and crowd-sourced databases – offers new opportunities to model and monitor urban built environment emissions. If properly processed and integrated, these data can support the development of standardized, simulation-ready building archetypes that enable citywide energy modeling (Ali, Shamsi, Hoare, Mangina, & O'Donnell, 2021). Open-source urban building energy modeling (UBEM) platforms hold immense potential: they democratize access to carbon performance data, foster reproducibility, and empower cities to design and evaluate context-specific retrofit strategies using a common digital framework.

Our study develops an open-source UBEM and carbon emissions framework that is scalable and replicable. The framework integrates geographical information systems (GIS) data, UBEM archetype templates, and result visualizations to support energy (and carbon emissions) analysis, policy evaluation, and public engagement. To demonstrate applicability, Zurich is provided as a baseline example in the GitHub repository, while Singapore is utilized as a comprehensive, city-scale proof of concept (instantiated as <https://buildings.sg/>). As a highly urbanized city with clear urban planning, the city-state offers unique research value – Singapore serves as an ideal proof of concept due to its urban density and year-round cooling demand, which rigorously tests our framework's scalability. Furthermore, by deploying the platform to model climate targets – such as Singapore's goal of 80% of new buildings meeting Super Low Energy Standards by 2030 (Building and Construction Authority, 2022) – we demonstrate how the toolkit can effectively support ambitious decarbonization masterplans in municipalities worldwide.

2. Background

2.1. Platforms for urban building energy models and carbon emissions maps

UBEM has emerged as a vital tool for urban sustainability, with research in this domain growing significantly since the late 2010s (Ali et al., 2021). By applying physics-based models of heat and mass transfer within and around buildings (Reinhart & Davila, 2016), UBEM enables simulations of current (and future) energy and carbon emissions performance at the neighborhood (Chen, Wang, Wong, & Ang, 2026) or city scale. These insights support sustainable urban planning, policy formulation, and targeted energy retrofits (Ali et al., 2021; Haneef, Pernigotto, Gasparella, & Kämpf, 2021; Hao & Hong, 2021; Song et al., 2025).

The development of a UBEM typically involves four key stages (Ang, Berzolla, & Reinhart, 2020; Davila, Reinhart, & Bemis, 2016; Zhang, Chen, Xu, & Wang, 2025): (a) Model characterization, which integrates data on building geometry, construction materials, physical properties, HVAC systems, and occupancy patterns. (b) Model generation, which translates these datasets into simulation-ready models for engines such as EnergyPlus. (c) Energy simulation, which estimates energy demand, peak loads, and emissions under specified climatic and operational conditions. (d) Visualization, which maps and summarizes results to support comparison, benchmarking, and scenario analysis. Table 1 compares representative UBEM platforms, highlighting their diverse modeling scales, input data structures, and analytical functions. Early tools such as UMI (Reinhart et al., 2013) were mainly developed for neighborhood-level to city-scale analysis, but requires Rhinoceros 3D as a CAD engine.

As the field matured, (more recent) platforms shifted toward urban-

Table 1

Summary of EnergyPlus-based bottom-up UBEM platforms.

Platform	Scale	Input Modules	Functions	Reference
UMI	Neighborhood	Building geometry (Rhinoceros 3D), urban context, building use / occupancy patterns	Building energy, daylighting, outdoor thermal comfort, and walkability assessment	Reinhart, Dogan, Jakubiec, Rakha, & Sang, 2013
CityBES	Urban	Weather file, building stock, GIS, database (ECMs, utility rates, etc.)	Energy retrofit analysis, building benchmarking, rooftop energy modeling, heat resilience modeling, district energy, urban mapping (UHI, HVI)	Hong, Chen, Lee, & Piette, 2016
CityEL	Urban	Building footprint, building archetypes, district boundaries, user settings, weather file	Building benchmarking, retrofit analysis, rooftop solar potential, comparative analysis	Song et al., 2025
GeoBEM	Urban	GIS data, building use / occupancy patterns, material, weather file, measured energy data, built year, building type, building height	Spatial distribution of energy consumption, urban building energy usage patterns analysis	Zhang et al., 2025

scale applications by integrating broader geospatial datasets. For example, CityBES (Hong et al., 2016) leverages building-stock information, GIS data, and databases to support district energy assessment and heat mapping. Similarly, CityEL (Song et al., 2025) introduces web-based workflows for retrofit evaluation and solar potential analysis. GeoBEM (Zhang et al., 2025) synthesizes diverse building attributes (e. g., age, materials) with measured data to map citywide energy consumption patterns.

On data and archetypes, the European TABULA project (Loga et al., 2012) and the U.S. Department of Energy (DOE) prototype buildings library (Deru et al., 2011) are among the most widely used sources of building characteristics and simulation templates. However, TABULA's datasets often (still) require significant manual transformation into simulation-ready formats. Conversely, while the DOE templates are comprehensive, they are primarily contextualized for the United States – they also require standalone simulation and lack integrated, visualization-ready outputs for downstream applications, especially policy planning. To bridge similar urban data gaps, recent studies have explored machine learning techniques applied to publicly available satellite imagery (Wang, Reinhart, & Ang, 2025) or the consolidation of disparate public records such as tax assessment parcels (Chen, Hong, Luo, & Hooper, 2019).

2.2. Research problem and gap

Despite the recent proliferation of UBEM studies, several key gaps persist:

1. Incompleteness assessment: Most existing tools focus primarily on operational energy, with limited consideration of embodied carbon,

- thereby failing to capture the full life-cycle energy and carbon impacts of urban building stocks (De Wolf, Pomponi, & Moncaster, 2017).
2. Limited accessibility and interactivity: While some web-based platforms – such as UBEM.io (Ang, Berzolla, Letellier-Duchesne, Jusiega, & Reinhart, 2022) and the Hong Kong UBEM visualization platform (Zhang et al., 2025) – have attempted to enhance accessibility, the majority of frameworks still produce static, non-interactive visualizations with limited data transparency and analytical depth, reducing their efficacy for scenario analysis, stakeholder engagement, and policy planning.
 3. Methodological and data constraints: Integrating diverse urban-scale datasets – including building geometry, envelope properties, mechanical systems, and occupancy patterns – remains technically challenging. The scarcity of openly available, locally validated archetype libraries (especially for tropical climates), severely limits the scalability of current models.

These gaps highlight the need for a modular, reproducible, and publicly accessible UBEM platform. Such a framework should seamlessly simulate and map both operational and embodied carbon across urban building stocks, support rigorous life-cycle assessments, and provide interactive, stakeholder-oriented visualizations to facilitate informed decision-making.

Compared with existing UBEM research and platforms, this study introduces three methodological contributions. First, it establishes an end-to-end UBEM workflow that integrates urban data preprocessing, machine learning-based archetype inference, EnergyPlus-ready archetype generation, and web-based visualization within a single reproducible pipeline. Second, the framework extends conventional UBEM

approaches (which focus primarily on operational energy) by jointly modeling operational and probabilistic embodied carbon at the city scale. To validate this capability, the study introduces comprehensive simulation templates that cover the entire building stock of the Singapore proof of concept. Third, the framework emphasizes reproducibility and transferability by relying on globally available open geospatial data (e.g., OSM) and openly releasing IDF templates and GeoJSON datasets, enabling the workflow to be adapted to other cities with minimal modification.

3. Methodology and technical implementation

The overall framework for this study (Fig. 1) comprises four stages: data collection and processing; machine learning for building archetype prediction; (embodied and operational) carbon emissions simulation; and development of the open platform (Buildings.city → Buildings.sg).

3.1. Data collection and processing

3.1.1. Data sources

The Buildings.city framework is designed to leverage globally accessible geospatial data, such as OpenStreetMap (OSM), which can then be augmented with specialized data from local municipal authorities to enrich building attributes. For the Singapore proof of concept, the initial data collection phase focused on building footprint geometries sourced from OpenStreetMap (OSM), supplemented with additional datasets from the Singapore Housing & Development Board (HDB) and Building and Construction Authority (BCA), as detailed in Table 2. The OSM dataset contains a total of 122,522 entries. The HDB dataset (Biljecki, 2020), with 12,119 entries, was matched to the OSM data

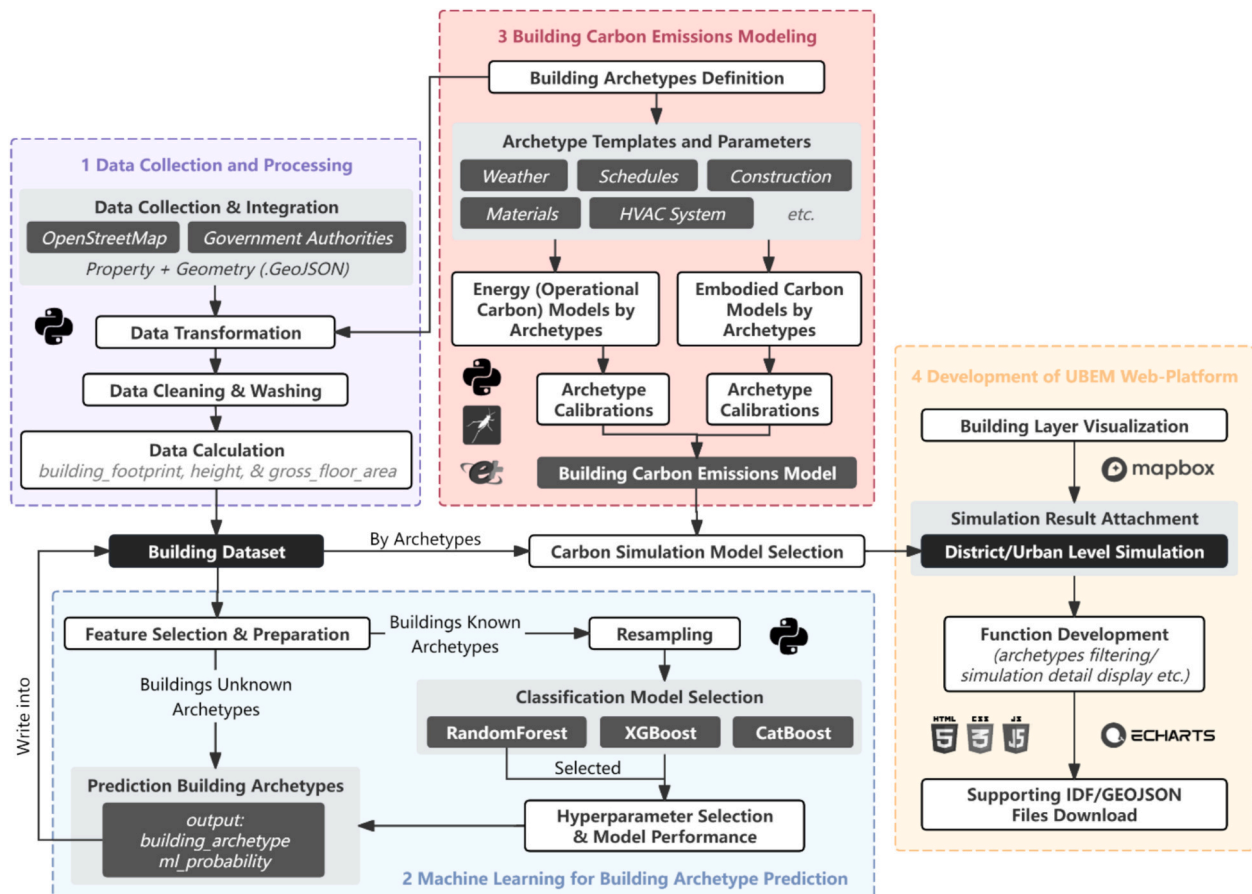


Fig. 1. Development of Buildings.city platform framework and simulation pipeline.

Table 2

Data source overview for the Singapore UBEM platform.

Data Source	Version	Original Fields (Key Attributes)	Format	Source Link
OpenStreetMap	2025-04-09	id, addr:housename, addr:housenumber, addr:postcode, addr:street, building, building:levels, height, type, coordinates	GeoJSON	https://overpass-turbo.eu
HDB	2019-08-25	hdb_max_floor_lvl, hdb_year_completed, hdb_total_dwelling_units, height	CityJSON	https://github.com/uasg/hdb3d-data
BCA (2024)	2024-12-24	Building Name, Building Address, Building Postal Code, Building Type, 2021 EUI, 2022 EUI, 2023 EUI, TOP/CSC Year, Total GFA of Building/Development (m2), Total Air-Conditioned Floor Area (m2), AC Floor Area Percentage (%), Type of AC System, Chilled Water System Efficiency (kW/RT), Air-side Efficiency (kW/RT)	XLSX	https://www1.bca.gov.sg/buildsg/sustainability/regulatory-requirements-for-existing-buildings/bca-building-energy-benchmarking-and-disclosure
BCA (2020)	2024-06-06	building type, year built, greenmark rating, greenmark year of award, greenmark version, gross floor area, percentage of airconditioned floor area, average monthly building occupancy rate	CSV	https://data.gov.sg/collections/22/datasets/d_e86d8a219d0936dbb321ade068a381da/view

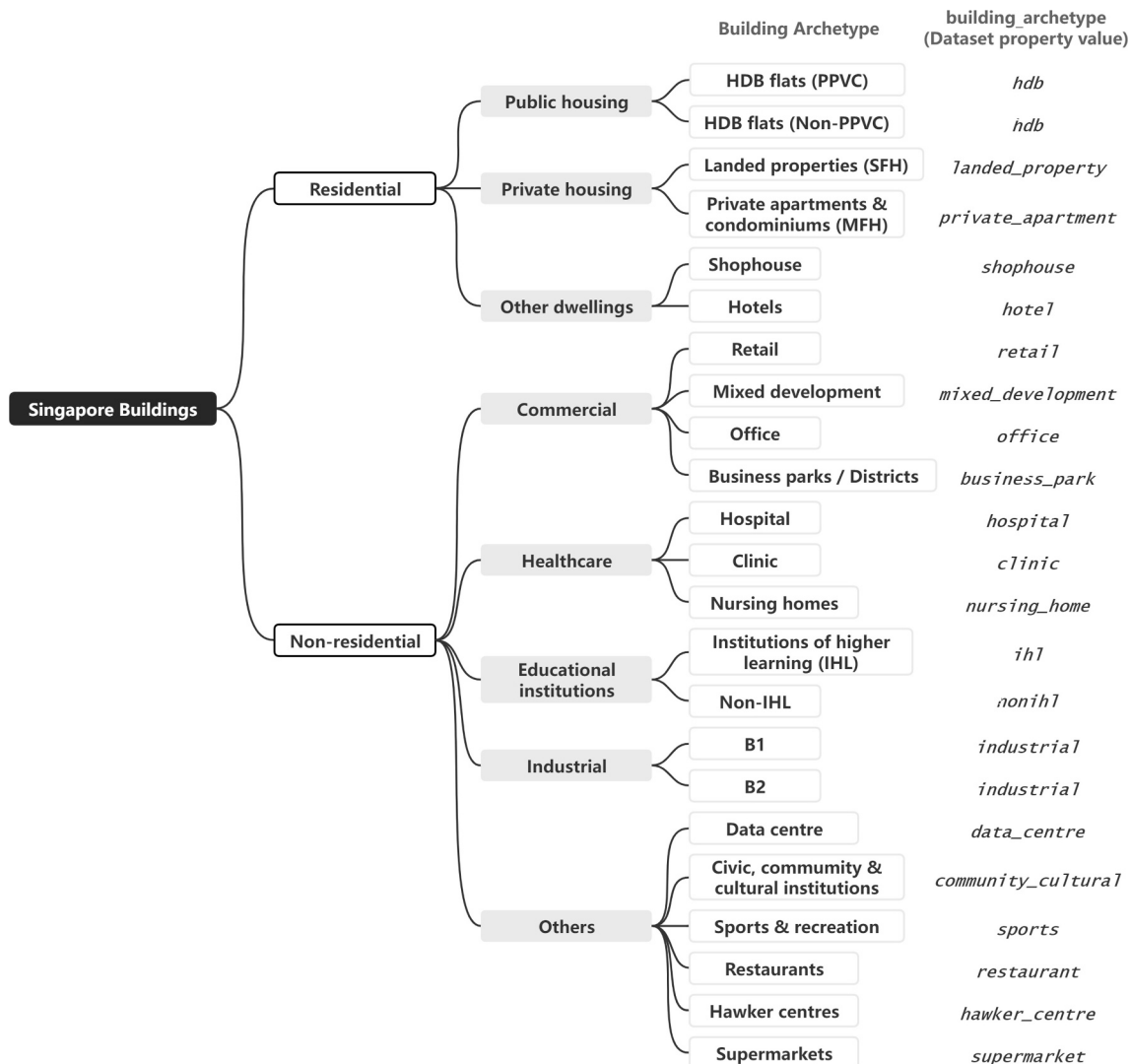
using unique identifiers. The BCA dataset, consisting of 1,005 entries (primarily for non-residential buildings), was linked through postal codes, building names, and addresses. The HDB and BCA datasets matched OSM at rates of 98% and 91%, respectively.

3.1.2. Data processing

The data processing pipeline includes data cleaning, classification of archetypes and height-related attributes, and calculation of building areas. For data cleaning, records were excluded if levels were non-

integer or the geometry type was neither “polygon” nor “multipolygon”, to remove implausible entries.

3.1.2.1. Building archetypes processing. Our framework requires the definition of context-specific building archetypes to ensure localized accuracy. As shown in Fig. 2 (for the proof of concept) the entire Singapore building stock was classified into 23 archetypes. This classification framework was developed in consultation with the relevant government agencies to ensure alignment with local planning standards

**Fig. 2.** Classification approach for building archetype categories in this study.

(see Section 3.3.1 for further details). Raw “building” attribute values in the OSM GeoJSON dataset were pre-processed to remove irrelevant categories such as bus stops, bridges, garages, and metro stations. Based on the cleaned dataset, building archetypes were inferred from the *addr:housename* field using two approaches. Functional terms such as “clinic” and “nursing home” enabled direct classification. For other categories, representative names – such as “McDonald’s” for *restaurant*, “Food Republic” for *hawker centre*, and “Gym” for *sports* – served as indicative examples (implicit categorization) based on brand recognition or semantic association. To ensure robust classification across the dataset, a comprehensive dictionary of over 50 keywords was compiled for each category.

Subsequently, values for the *building archetype* field were partially assigned based on HDB and BCA data – this source data was prioritized over OSM due to its accuracy and local relevance. However, while OSM provides (relatively) extensive footprint coverage, it suffers from significant spatial heterogeneity and attribute incompleteness, particularly regarding semantic building information (Bernard, Bocher, Le Saux Wiederhold, Leconte, & Masson, 2022; Biljecki, Chow, & Lee, 2023; Hecht, Kunze, & Hahmann, 2013). In this study, around 64% of building functional types remained unknown. Filling these semantic data gaps in crowdsourced platforms has been a major focus in urban modeling research for quite some time. Previous studies have addressed this through various methods, including spatial interpolation, rule-based inference heuristics, and the integration of auxiliary datasets such as remote sensing or LiDAR (Barranquero et al., 2023). Building upon this foundation, a machine learning approach is developed to predict unknown building archetypes (Section 3.2).

3.1.2.2. Building height and levels processing. Analysis of the OSM data revealed extensive missing data in building heights and storey count fields. Even after applying a standard estimation of three meters per storey, over 72% of entries lacked valid values. The most significant gaps were observed for “landed properties” and “industrial buildings”, with 15,339 (82%) and 1951 (71%) incomplete entries, respectively. Smaller building typologies, such as “sports” and “supermarket” facilities, also exhibited critical data gaps, with each exceeding 80% missing height data. In contrast, the HDB dataset showed markedly higher completeness, with less than 5% of entries missing height values.

Notably, the missing “landed property” data exhibited spatial clustering. To mitigate this, a road-name-based validation was conducted using Google Earth’s 3D measurements and street view imagery, yielding complete height data for 2,796 landed properties across 26 representative communities. Despite these efforts, certain classification challenges persist, particularly in distinguishing Prefabricated Prefinished Volumetric Construction (PPVC) from non-PPVC HDB buildings, and in differentiating B1 from B2 industrial facilities.

For spatial visualization of energy simulation results, building footprint areas were calculated using GeoJSON polygon coordinates. Structures with an area $\leq 50 \text{ m}^2$ were excluded as likely non-primary buildings. Total floor area was estimated by multiplying the footprint area by the number of storeys. When height data remained unavailable, a conservative baseline of a single storey was assumed (solely) for visualization purposes. Following these processing steps, the final cleaned dataset contained 118,782 valid building entries.

3.2. Machine learning for building archetype prediction

Due to substantial label gaps in the consolidated datasets, we trained a supervised learning model using the numerical and geometric features derived from spatial data.

3.2.1. Feature engineering and selection for predictive modeling

The target variable in the prediction model was the categorical variable *building archetype*, with 23 distinct classes. All input features

were normalized before training to ensure comparability and prevent gradient distortions. The numerical features in our model included *gross_floor_area*, *building footprint*, *height*, and *building levels* (i.e., predicting building archetype or functional use primarily based on morphological characteristics). The *building footprint* feature represents the area of each building’s footprint. The *built_year* attribute was excluded due to its low feature importance and high missing rate, a decision aligned with prior practices to reduce noise and enhance generalization (Guyon & Elisseeff, 2003).

To further characterize spatial morphology, building (polygon) geometries were transformed into interpretable shape descriptors using the Shapely library. These derived features included *perimeter*, *aspect_ratio*, *compactness_ratio*, and *convexity_ratio*. These shape-based features capture the distinct “morphological signatures” of different urban building typologies. For instance, metrics such as compactness and convexity effectively differentiate porous institutional complexes (e.g., Institutes of Higher Learning) from more monolithic or uniform residential blocks.

3.2.2. 5-fold cross-validation

Previous studies (James, Witten, Hastie, & Tibshirani, 2013; Kohavi, 1995) have demonstrated that k-fold cross-validation provides a more reliable estimate of model performance in small-sample tasks compared to a single split (e.g., 80/20), as it maximizes data utilization during evaluation. In this study, the primary evaluation metric was the 5-fold cross-validated accuracy, calculated as:

$$\text{cross_val_score} = \bar{a} \pm \sigma_a, \quad \bar{a} = \frac{1}{K} \sum_{i=1}^K a_i, \quad K = 5 \quad (1)$$

This captures both the central tendency and variability of model performance across folds, providing a more reliable estimate of generalization than single-split evaluations, especially for limited or imbalanced datasets (Raschka, 2020). Applying oversampling to the entire dataset before a single 80/20 train-test split can introduce data leakage, because synthetic samples may duplicate or closely resemble instances in the test set. This can lead to artificially inflated performance metrics (e.g., precision = 1, log loss = 0), as we observed when using the XGBoost model. By performing oversampling within each fold after k-fold cross-validation, the model is evaluated on unseen data in each fold, thereby preserving the integrity of performance metrics (Lemaître, Nogueira, & Aridas, 2017).

3.2.3. Resampling to mitigate class imbalance

The distribution of building archetypes in the training dataset ($n = 42,648$) exhibited significant class imbalance, with a few dominant categories (particularly HDB) comprising the majority of samples. To prevent model bias toward majority classes and mitigate the risk of overfitting, a hybrid resampling strategy was implemented. This approach combines both undersampling of majority classes and oversampling of minority classes. Specifically, the two largest classes (with over 10,000 samples) were undersampled to 80% of their original sizes to reduce their disproportionate influence. For the remaining under-represented classes, the target sample size was established as 20 times the original count. When the target exceeded ten instances, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to generate synthetic samples. For smaller categories, random oversampling was employed to minimize the risk of introducing excessive noise (Blagus & Lusa, 2013). This multi-stage, balanced resampling procedure ensured that the model was trained on a more uniform class distribution while preserving generalization capabilities.

3.2.4. Supervised classification model selection

We then evaluated three supervised learning models (viz., Random Forest, XGBoost, and CatBoost) for predicting building archetypes in Singapore. Their cross-validated performance scores and feature importance rankings are presented in Fig. 3.

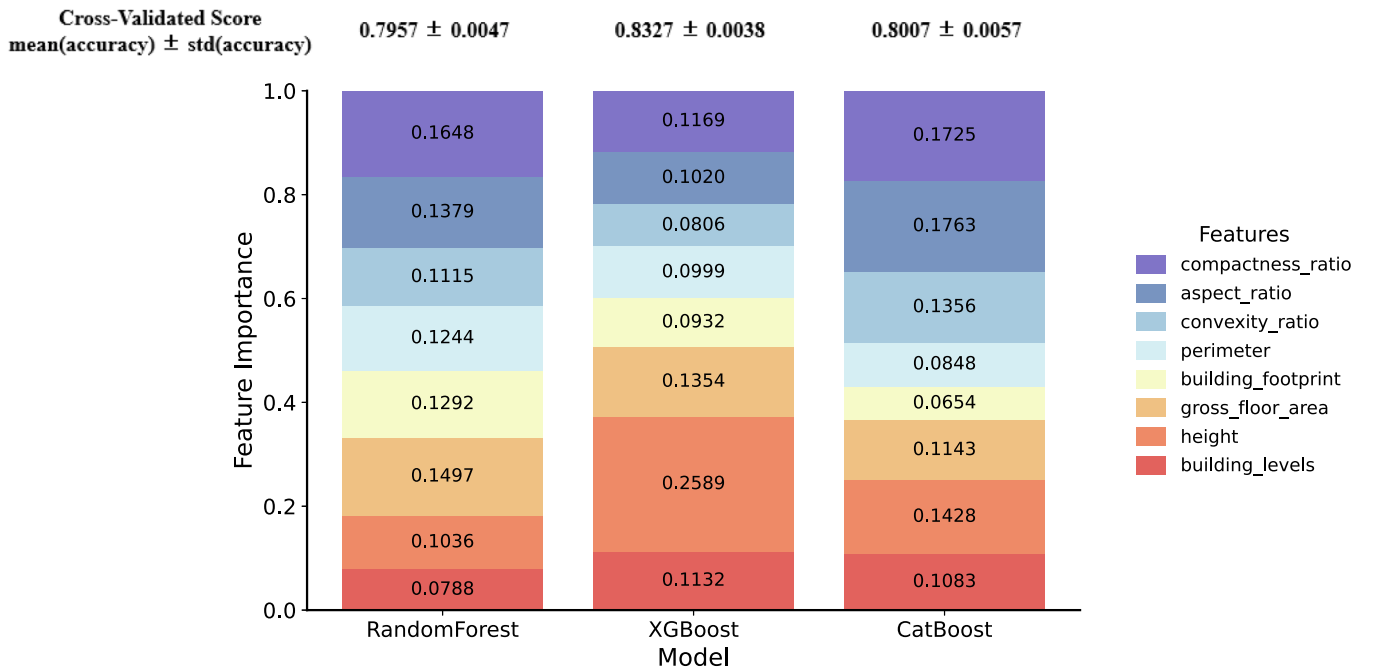


Fig. 3. Accuracy and feature importance comparison of Random Forest, XGBoost, and CatBoost.

Among the evaluated models, XGBoost demonstrated the highest predictive accuracy and stability, achieving a mean accuracy of 83.27% and the lowest standard deviation (0.0038) across cross-validation folds. CatBoost and Random Forest achieved slightly lower accuracies of 80.07% and 79.57%, respectively, and showed marginally higher variance across folds.

These models estimate feature importance using distinct algorithms, enabling meaningful cross-model comparisons (Breiman, 2001; Chen & Guestrin, 2016; Prokhorenkova, Gusev, Vorobev, Dorogush, & Gulin, 2018; Wang, Liang, Hancock, & Khoshgoftaar, 2024). Notably, XGBoost exhibited a disproportionately high reliance on *height* (25.89%), suggesting potential over-reliance on (and potential sensitivity to errors in) the height features. In contrast, CatBoost prioritized *aspect_ratio* (17.63%) and *compactness_ratio* (17.25%), indicating greater sensitivity to geometric descriptors. Random Forest, however, displayed the most equitable distribution of feature importance across the input vector, indicating enhanced resilience against biases inherent in any single variable.

Based on these trade-offs, Random Forest was ultimately selected as the final model for building archetype prediction. Although its mean accuracy was slightly lower than that of boosting-based models, it demonstrated superior stability and generalization potential. Crucially, its (more uniform) feature importance suggested reduced dependence on *height*, *building_levels*, and *gross_floor_area* – variables that, in the context of OSM data, often contain the highest rate of missing or derived values. These characteristics render Random Forest a more robust choice under imperfect data conditions.

3.2.5. Hyperparameter optimization

Hyperparameter tuning is essential for enhancing the predictive performance and generalizability of Random Forest models. Common techniques include grid search, random search, and Bayesian optimization (Feurer & Hutter, 2019; Yang & Shami, 2020). The study adopted a stepwise optimization strategy: first fixing the number of trees ($n_{estimators}$) to ensure stability, then fine-tuning structural parameters such as max_depth , $min_samples_split$, and $min_samples_leaf$ to control model complexity and reduce overfitting. Model performance was assessed using five-fold cross-validation. Metrics, including accuracy, F1 score,

and log loss, were reported for both training and testing sets to support a comprehensive evaluation and to detect overfitting through comparative performance analysis.

Literature suggests that Random Forest performance typically stabilizes beyond roughly 100 trees (Lange, Gültas, Schmitt, & Heinrich, 2025). Increasing $n_{estimators}$ can marginally improve accuracy, it also incurs higher computational cost with diminishing returns in generalization (Feurer & Hutter, 2019; Lange et al., 2025). In our tuning, accuracy and log loss converged at $n_{estimators} = 500$, which was established as the baseline for subsequent optimization. To further fine-tune model complexity, a grid search was conducted over the hyperparameters max_depth , $min_samples_split$, and $min_samples_leaf$ (Bischl et al., 2023). The selected parameter ranges, shown in Table 3, were determined through preliminary trials to prevent both underfitting and overfitting.

The evaluation results of models with different hyperparameter combinations on the test set are shown in Fig. 4. The model's performance metrics tend to stabilize when the maximum tree depth approaches 20, whereas a depth of 10 resulted in clear underfitting. Smaller values can increase accuracy but also heighten the risk of overfitting, as evidenced by greater variance in cross-validation scores. To balance the bias-variance trade-off (and to obtain a more robust classification model), we selected the hyperparameter combination with the smallest cross-validation score variance (0.00389) within the maximum depths of 20 to 30, specifically: $n_{estimators} = 500$, $max_depth = 20$, $min_samples_split = 10$, and $min_samples_leaf = 4$. This configuration achieves a high mean accuracy (0.781), a high F1-score (0.805), and a low log loss (0.792).

Table 3
Hyperparameter grid search space.

Hyperparameter	Type	Search Space
$n_{estimators}$	Discrete	[500]
max_depth	Discrete	[10, 15, 20, 30, 50]
$min_samples_split$	Discrete	[5, 10, 15, 20]
$min_samples_leaf$	Discrete	[2, 4, 6]

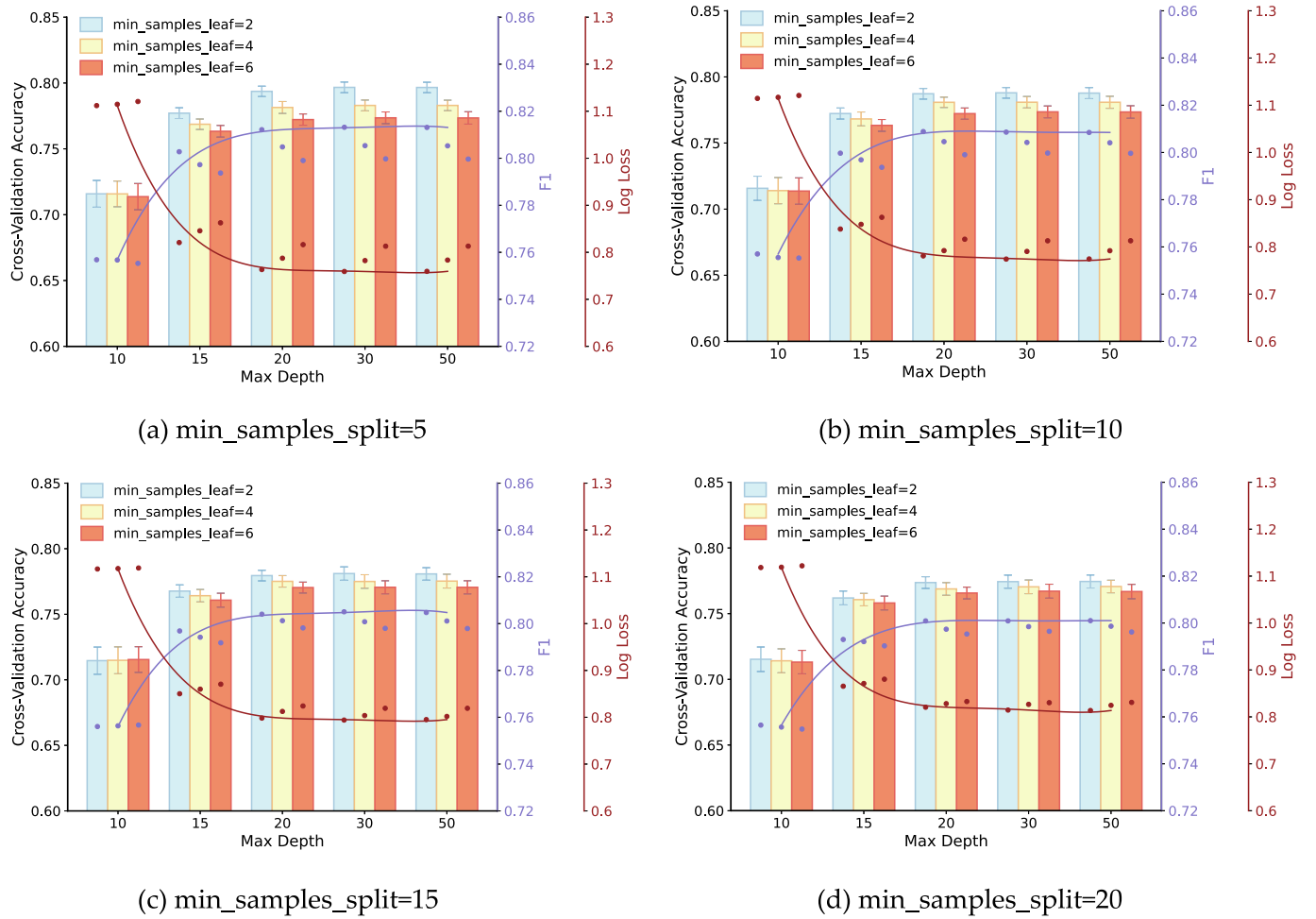


Fig. 4. Hyperparameter tuning and performance for building architecture prediction.

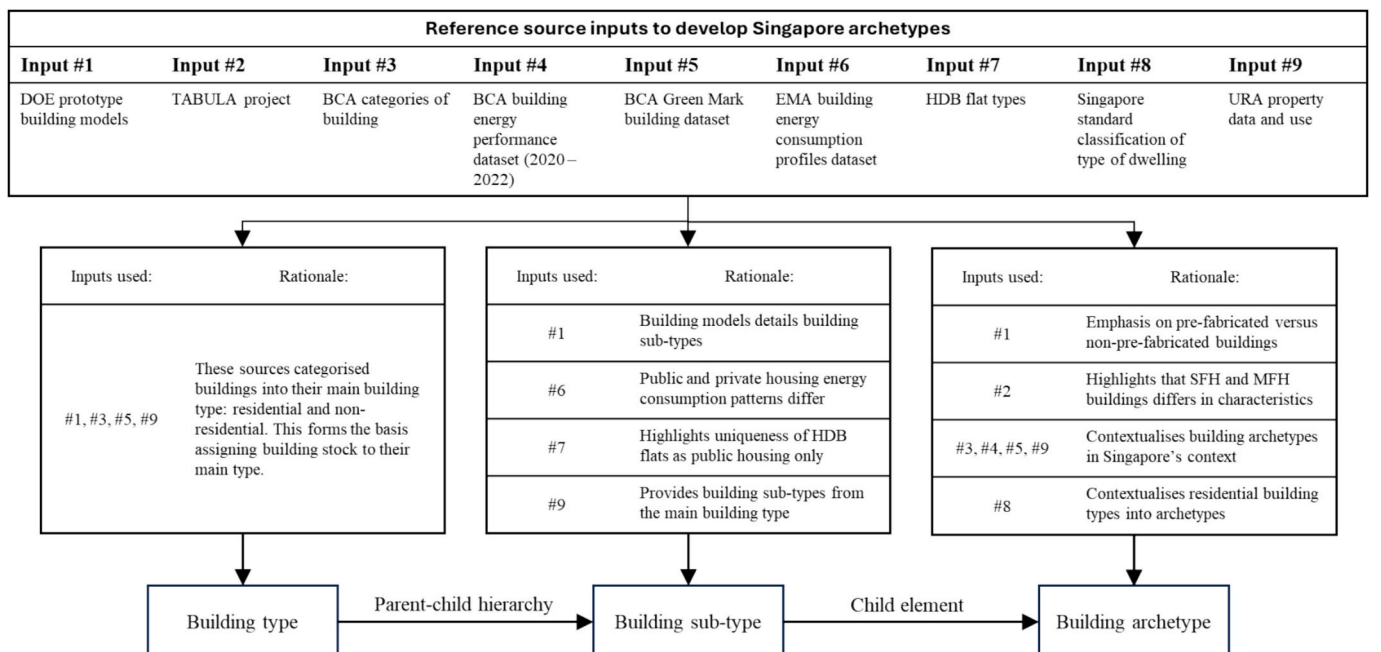


Fig. 5. The (source) inputs and processes for Singapore building archetype classification.

3.3. Building carbon simulations for archetypes

3.3.1. Archetype classification

A critical challenge in global UBE deployment is the lack of standardized building archetypes, as individual municipal agencies often use fragmented classification systems. To address this, the Buildings.city framework employs a hierarchical classification methodology that harmonizes international standards – such as the U.S. DOE prototype models and the TABULA project (covering multiple EU countries) – with local municipal data.

In the Singapore proof of concept, this approach was used to overcome local institutional fragmentation, as individual government agencies oversee different building types. Archetypes were developed by aggregating local practices from agencies including the Singapore Building and Construction Authority (BCA), Energy Market Authority (EMA), Housing and Development Board (HDB), Urban Redevelopment Authority (URA), and Department of Statistics (DOS). As illustrated in Fig. 5, the resulting application establishes a standardized repository of archetypes for the city-state. The classification hierarchy is bifurcated into two primary categories: residential and non-residential. These are further disaggregated into specific functional subtypes derived from consolidated building-stock data. Finally, these sub-types were mapped to energy-modeling archetypes, ensuring cross-agency consistency by harmonizing multiple local classification systems. This unified approach not only enhances the reproducibility of modeling but also provides a common, interoperable repository of archetypes suitable for multi-agency urban planning and policy evaluation in any municipality.

3.3.2. Building operational energy (and carbon emissions) modeling

3.3.2.1. Archetype characteristics and parameters. Energy modeling templates were developed for each building archetype as EnergyPlus input data files (.idf), incorporating both geometric and non-geometric parameters required for parametric energy simulation. All templates for the Singapore proof of concept are publicly accessible on <https://Buildings.sg> or GitHub (see Code and Data Section), with key parameters (viz., construction materials, glazing systems, and window-to-wall ratios) derived from local building codes, standards, and guidelines issued by relevant Singaporean authorities.

While geometric attributes were informed by spatial data, non-geometric parameters required additional assumptions and adaptations. Default occupancy schedules available in simulation software

libraries are often based on norms and may not accurately reflect local building use patterns. To address this, custom occupancy schedules were created to more accurately reflect real-world conditions. Instead of relying on resource-intensive survey methods to construct such schedules, a data-driven approach was adopted. Specifically, Google Maps' "Popular Times" feature (which provides hourly footfall estimates normalized between 0 and 1 for buildings with substantial visitor traffic) was leveraged. Using an open-source extraction tool, time-series data were collected from representative buildings for each archetype where "Popular Times" data were available. These profiles were then translated into occupancy schedules used in the EnergyPlus models, providing a scalable and context-sensitive approximation of building usage behaviour (Fig. 6).

3.3.2.2. Energy models. The platform uses EnergyPlus as its core simulation engine. As shown in Fig. 7, a representative energy model was developed for each archetype using a "shoebox" abstraction to represent building storeys. This geometric approach is designed to facilitate rapid, large-scale simulations with minimal computational overhead. Compared to high-fidelity semantic 3D models, the shoebox method offers superior performance for urban-scale modeling by enabling efficient parameterization, optimization, and sensitivity analyses. Research indicated that at the city scale, the marginal accuracy gains provided by detailed energy models are outweighed by increased complexity of data acquisition and processing (Battini, Pernigotto, & Gasparella, 2021; León-Sánchez, Agugiaro, & Stoter, 2025; Widl, Agugiaro, & Puerto, 2018).

The model geometry is divided into core and perimeter zones following ASHRAE energy modeling standards (ASHRAE, 2018, 2022a, 2022b, 2023) to account for differences in heat gains. Perimeter zones are primarily influenced by solar radiation and external weather conditions, whereas core zones are mainly affected by internal gains from lighting, equipment, and other internal loads. In the unit shoebox model, the perimeter zone is defined with a 1-m offset from the building edge, and the shoebox is standardized to a $10 \times 10 \times 3$ m volume. When applied to actual buildings, these zones are proportionally (and dynamically) scaled according to the building's real dimensions – for example, a $40 \times 40 \times 3$ m building would be simulated with a proportional 4 m perimeter depth. This scaling logic ensures that the thermal balance between exterior-influenced and interior-load-dominated spaces is accurately maintained across diverse building sizes.

These energy models were subsequently translated into simulation-

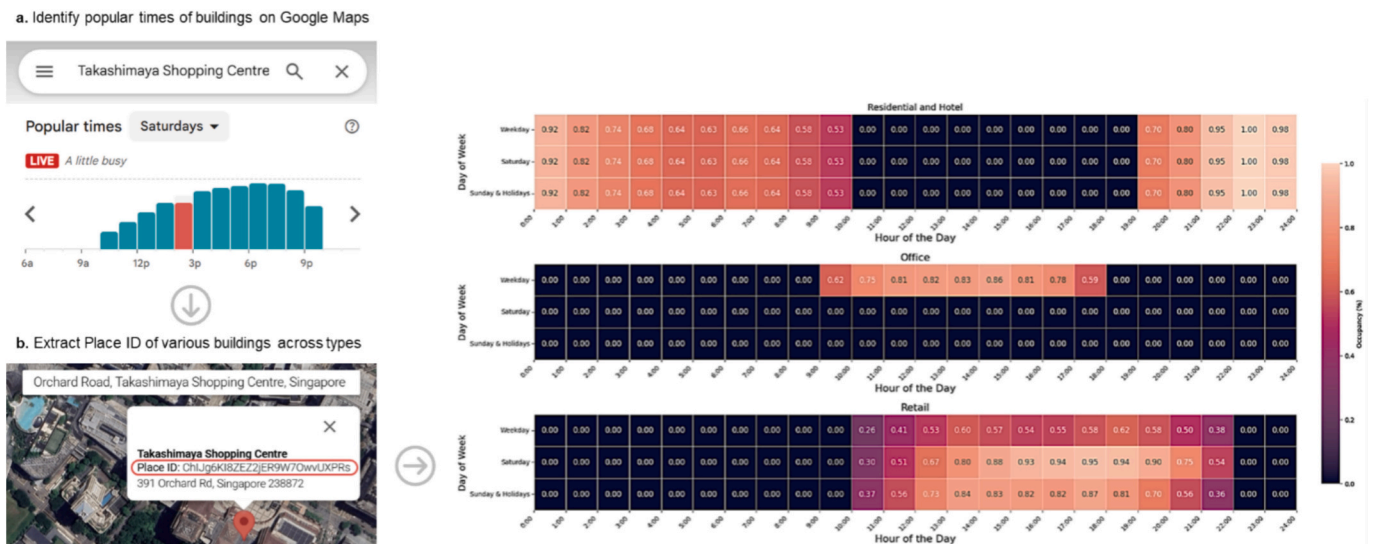


Fig. 6. Approach for deriving occupancy, lighting, and equipment schedules in the energy simulation templates and models. Spatial and functional attributes informing these schedules were accessed via the Google Maps Platform Places API (Place ID).

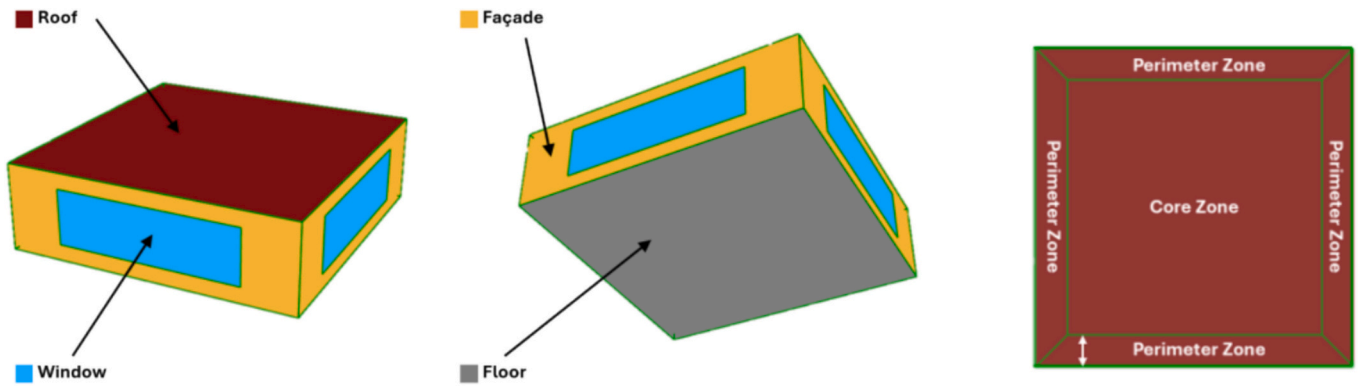


Fig. 7. Energy models for each archetype, with each building level as a single ASHRAE 5-zone shoebox.

ready states using Rhinoceros 3D and Grasshopper, with the integration of Ladybug Tools (an open-source suite of environmental design plugins for building performance simulation). Specifically, the Honeybee and Honeybee-Energy components were employed to define and assign key building characteristics within the modeling environment, while Ladybug provided climate data integration and visualization capabilities. Following the template setup, initial simulations were conducted for calibration purposes. This step involved comparing predicted energy consumption outputs against reported performance data or established benchmark values to validate model accuracy and ensure reliability prior to conducting scenario-based analyses. All simulations were conducted using a validated Typical Meteorological Year (TMY) file for Singapore from 2001 to 2020 in the EnergyPlus Weather (EPW) file format.

3.3.2.3. Archetype operational calibration. Publicly available energy data were used to benchmark the model results (outputs). For the proof of concept, the Singapore BCA publishes annual EUI data based on self-disclosed information from building owners. Distributions were developed for each archetype for benchmarking or “partial calibration” (given the lack of monthly or higher-fidelity energy datasets) and were used to assess the consistency of simulation outputs. The representative archetypes underwent an iterative adjustment process to ensure that the simulated results remained within a reasonable Normalized Mean Bias Error (NMBE) range. The NMBE was applied as a statistical indicator of bias between simulated and reference EUI values. The equation for NMBE is:

$$NMBE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{(n - p) \times \bar{y}} \quad (2)$$

Where: n = number of data points, y_i = actual measured EUI of building i , $\hat{y}_i$ = predicted EUI of building i , and $\bar{y}$ = mean of the actual measured EUI of n buildings.

Across the primary archetypes (including hotel, retail, office, hospital, clinic, nursing home, IHL, civic institution, and sports and recreation), the models achieved NMBE values within $\pm 10\%$. This indicates strong agreement with benchmark data and demonstrates sufficient consistency for baseline operational energy and carbon assessments. Given that only annual data were available, this approach serves as a robust cross-sectional validation rather than a high-fidelity time-series calibration.

Due to limitations in publicly available datasets, direct comparison with measured performance for the selected archetypes was not always feasible. For archetypes where NMBE was not available, the predicted EUIs were benchmarked against secondary reference sources, such as published standards and prior modeling studies. Archetypes with predicted EUIs within $\pm 20\%$ were considered acceptable for preliminary UBEM validation. The equation for benchmarking is given below:

$$\text{Difference from benchmark} = \frac{(\text{Predicted EUI} - \text{Benchmark EUI})}{\text{Benchmark EUI}} \times 100\% \quad (3)$$

3.3.3. Building embodied carbon modeling

3.3.3.1. Embodied carbon modeling scope definition. The proposed embodied carbon quantification method builds on widely accepted LCA frameworks, including the IStructE module definitions (Gibbons, Orr, Archer-Jones, Arnold, & Green, 2022). Considering data limitations and future uncertainties, a full LCA covering all modules and materials is impractical. Therefore, the scope was confined to modules A1–A5 (Fig. 8), which represent near-term, upfront carbon emissions critical to mitigating global warming. Five key materials were prioritized (viz., concrete, steel, glass, aluminum, and bricks) based on regulatory relevance and high prevalence in typical building stock. While other modules and materials were excluded for feasibility reasons, the selected scope captures most of the upfront carbon and ensures sufficiently comprehensive results for urban-scale assessments, where some errors at the individual building level are acceptable. Such a trade-off is widely accepted in urban-scale simulations and analysis, where the objective is to identify citywide trends and hotspots rather than achieve high-fidelity precision for individual structures (Hong et al., 2020; Reinhart & Davila, 2016).

3.3.3.2. Material data and emission factors. Estimating embodied carbon at the building level requires comprehensive data across multiple categories, processed to account for uncertainties in material quantities, source origins, and emission factors. Material quantities were expressed as intensities (kg/m^2 of GFA) and sourced from authoritative databases and peer-reviewed literature (Fishman, Mastrucci, Peled, et al., 2024). Each data point was classified according to archetype-material combinations (for example, glass in multi-family housing). Combinations with sufficient observations ($n \geq 8$) were fitted to probability density functions (PDFs) using SciPy, whereas combinations with limited data relied on mean values to ensure representative estimates. For the proof of concept, given Singapore’s import-dependent construction sector, all materials were assumed to originate overseas. Import shares by country were derived from UN Comtrade data (United Nations, 2025) and used to construct probability mass functions (PMFs) representing the likelihood that each material is sourced from specific countries.

Emission conversion factors (ECFs) quantifying the carbon impact of material extraction and manufacturing (A1–A3) were obtained from peer-reviewed studies and Environmental Product Declarations (EPDs). ECFs were categorized by material-country combinations, with PDFs or mean values determined following the same approach as material intensities. Transport emissions (A4) were calculated using mean shipping distances from each source country, derived from industry resources,

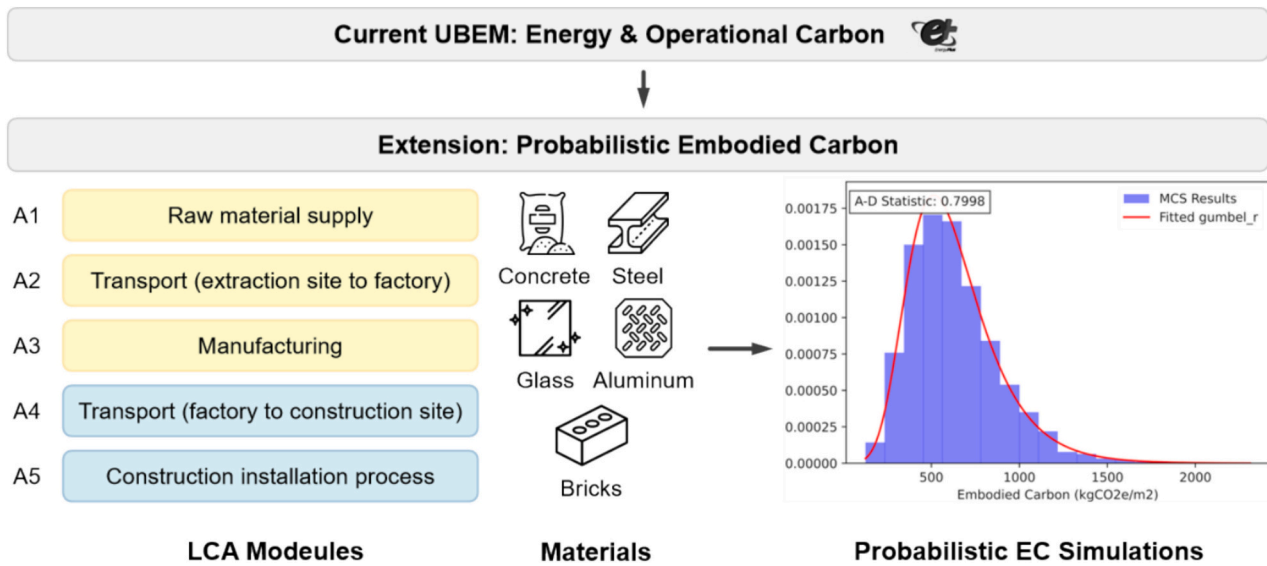


Fig. 8. UBE Model augmented with probabilistic embodied carbon simulation.

combined with published transport emission factors ($\text{kgCO}_2\text{e/kg/km}$) to produce A4 ECFs.

Material wastage and on-site emissions (A5) were incorporated using mean waste rates from industry guidelines, which were converted into waste factors. On-site construction emissions ($\text{kgCO}_2\text{e/m}^2$) were extracted from the literature and fitted to probability distributions consistent with material quantity treatment (Gibbons et al., 2022). By integrating uncertainties across material quantities, sourcing, manufacturing, transport, and on-site processes, this methodology provides a robust, probabilistic basis for assessing building-level embodied carbon.

3.3.3.3. Model calibration and district-level embodied carbon assessment.

A weighting system was applied to A1–A3 ECFs, which dominate embodied carbon contributions. During calibration, the weighting was adjusted iteratively to satisfy a predefined error threshold while minimizing distortion of the original data. Benchmark values (EC_{bench}) were obtained from peer-reviewed studies and the OneClickLCA demonstration database. Due to the limited availability of archetype-specific benchmarks, a composite mean was used. The simulation output (EC_{sim}) was computed as the mean of each archetype and then aggregated across all archetypes. Calibration error was expressed as the percentage deviation between the benchmark (EC_{bench}) and the simulated aggregate (EC_{sim}). Iterative adjustments continued until a 4.82% error was achieved, below the 5% maximum threshold established in prior studies (Sokol, Davila, & Reinhart, 2017).

$$\text{Calibration error} = \text{Percentage error} = \frac{|EC_{\text{bench}} - EC_{\text{sim}}|}{EC_{\text{bench}}} \times 100\% \quad (4)$$

District-level embodied carbon was quantified through 5,000 Monte Carlo simulations. In each simulation, embodied carbon intensities were sampled from the probability distributions of each building's archetype and multiplied by its GFA. The values were then summed across all buildings to yield total district-wide embodied carbon for each run, providing a probabilistic assessment of the district's embodied carbon.

3.4. Development of the open platform

3.4.1. Technology stack

The core Buildings.city web platform was developed as an UBE Model and carbon-mapping tool using a modern web stack (JavaScript, HTML, and CSS). The mapping interface leverages the Mapbox GL JS framework,

which enables accelerated rendering of large-scale vector tilesets. Simulation results were injected directly into the geospatial metadata of the building footprints, allowing for real-time spatial querying and 3D rendering via the Mapbox API. To ensure seamless performance across multiple zoom levels (10–20), the city-scale GeoJSON datasets were processed using Tippecanoe, converting raw vector data into the MBTiles format for efficient stream-loading. The application architecture uses a modular data structure that supports dynamic attribute-based filtering and archetype color mapping, significantly enhancing the maintainability and future extensibility of the codebase. ECharts was integrated to provide reactive, high-dimensional data visualizations and dashboarding capabilities.

3.4.2. Key features

Building upon this technical foundation, the platform integrates several interactive modules designed to facilitate data-driven analysis and informed decision-making at the urban scale:

1. Multi-dimensional 3D visualization: Users can toggle between layers representing building archetypes, embodied carbon emissions, operational carbon emissions, and green building-related attributes.
2. Archetype-specific analytics: A filtering system allows users to retrieve the specific simulation methodologies, input parameters, and aggregated results for each building category.
3. Data interoperability and open access: To support the broader research community, the platform provides export functionality for GeoJSON datasets, fully editable EnergyPlus IDF templates for all 23 archetypes (both detailed and simplified models). This enables users to perform localized sensitivity analyses or run customized simulations in external or local environments.
4. Granular building inspection and benchmark interface: A "click-to-inspect" interface provides immediate access to building-level metadata, including simulated EUI. A separate benchmarking tab allows building owners to benchmark their buildings and reported/published data.
5. Model transparency: Integrated toggle controls allow users to isolate buildings whose parameters were derived from machine learning-based archetype inference, enabling a transparent assessment of model-driven versus empirical data.

Together, these features provide an accessible, spatially contextualized environment for exploring city-scale energy and carbon performance, effectively bridging the gap between raw data and actionable

urban policy.

4. Results

4.1. Building archetype predictions

In the proof of concept, predictions were generated for all buildings in Singapore with unknown building archetype labels ($n = 76,134$) using the Random Forest model. The model's output probabilities ($ml_probability$) ranged from 0.1246 to 0.9999, with a mean of 0.5519 and a standard deviation of 0.2105. A Gaussian Mixture Model (GMM) was fitted to the predicted probabilities using the *GaussianMixture* class in scikit-learn, with a tied covariance structure and random initialization. The number of components (k) was selected by comparing the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for $k = 1$ to 5 (Table 4). The results indicate that moving from $k = 1$ to $k = 2$ substantially improves model fit, with a $\approx 48\%$ reduction in AIC/BIC. Further increases beyond $k = 2$ offer progressively smaller improvements, justifying the adoption of a two-component GMM to capture the dominant bimodal structure.

The fitted GMM components correspond to different levels of model confidence: the first peak (centered at 0.4115, weighted 0.6032) represents lower confidence, likely associated with samples near class boundaries with greater classification uncertainty (e.g., office and mixed-use developments); the second peak (centered at 0.7652, weighted 0.3968) indicates higher confidence, suggesting a stronger association with a specific *building archetype*. The probability density function (PDF) of the two-component GMM is expressed as:

$$p(x) = \sum_{k=1}^2 \pi_k \mathcal{N}(x | \mu_k, \sigma_k^2) \quad (5)$$

$$\mathcal{N}(x | \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right) \quad (6)$$

where the parameters are:

$$p(x) = 0.3968 \mathcal{N}(x | 0.7652, 0.0144) + 0.6032 \mathcal{N}(x | 0.4115, 0.0144) \quad (7)$$

Predicted probability distributions by building archetypes are illustrated in Fig. 9. In Fig. 9b, the distribution is relatively uniform, with the first and second peaks at similar levels. In Fig. 9c through 9e, the second peak gradually diminishes and shifts closer to the first peak. Notably, distribution Fig. 9f, corresponding to the 'retail' archetype, exhibits an almost unimodal distribution dominated by a single GMM peak. These patterns reflect how the Random Forest model differentiates building archetypes with varying levels of certainty: industrial properties and HDBs are more easily distinguishable, whereas offices, retail buildings, private apartments, and condominiums are more frequently misclassified.

4.2. Simulation results

4.2.1. Energy simulation

As shown in Fig. 10, baseline simulation results reveal clear differences in operational energy use and carbon emissions across building

Table 4

Model selection metrics for Gaussian Mixture Models (AIC and BIC values for $k = 1$ to 5).

k	AIC	BIC	Δ AIC vs $k = 1$	Δ BIC vs $k = 1$
1	-21,194	-21,176	–	–
2	-31,359	-31,322	10,164 ($\approx 48\%$)	10,146 ($\approx 48\%$)
3	-33,419	-33,364	2,060 ($\approx 7\%$)	2,042 ($\approx 6\%$)
4	-34,763	-34,689	1,344 ($\approx 4\%$)	1,325 ($\approx 4\%$)
5	-35,377	-35,285	614 ($\approx 2\%$)	596 ($\approx 2\%$)

archetypes in the proof of concept. Data centres exhibit the highest baseline (approximately 2,000 kWh/m²·yr, primarily due to intensive computational activity and high cooling demands), followed by supermarkets and Industrial B2 buildings. For most non-residential buildings, cooling clearly dominates energy consumption, accounting for 51–83% of total EUI. Equipment use is typically the second-largest contributor, particularly in industrial buildings, where data centres and Industrial B2 buildings have equipment energy shares exceeding 41% of total EUI. Water consumption is generally the smallest contributor, lower than lighting energy use – except in restaurants and hawker centres, where water use is relatively higher. Residential buildings (including Landed Properties, Private Apartments, and HDBs) have the lowest baseline energy consumption, with EUI ranging from 40 to 47 kWh/m²·yr. EUI for Non-IHL buildings is also relatively low, at 53.17 kWh/m²·yr. These lower values reflect reduced cooling and equipment energy demands, with HVAC cooling accounting for 37–44% of total EUI. This comparison underscores the importance of developing archetype-specific energy and carbon reduction strategies that account for differences in baseline performance.

4.2.2. Embodied carbon simulation

Three district-level scenarios (Fig. 11) were analyzed to explore variations in embodied carbon emissions under different assumptions, based on prior work by the authors in Soh, Wong, and Ang (2025). The first scenario serves as a baseline, using the original dataset without modification. The second scenario, "Asia-only imports," represents potential material sourcing patterns within the ASEAN free trade framework and current geopolitical context. The third scenario considers the adoption of prefabrication and modular construction (PAMC) technologies, which have been reported to reduce A1–A3 emissions by about 40% and material waste by at least 20% (Ding et al., 2024; Greer & Horvath, 2023). Results are expressed in kgCO₂e/m² of GFA.

The median embodied carbon of hotels is notably higher than that of other archetypes, indicating that hotels generally have higher embodied carbon and therefore warrant greater attention in emission reduction efforts. Meanwhile, the interquartile ranges (IQRs) for hotels, clinics, and sports and recreation buildings are considerably narrower, suggesting greater certainty and less variation in their predicted embodied carbon, whereas HDBs and private apartments exhibit wider IQRs. Sourcing materials exclusively from Asian countries has little effect, and in some cases, may even increase embodied carbon. For mixed development, both the median and IQR increased under this scenario, indicating that supply chain choices can have counterintuitive effects. The PAMC scenario shifts the distributions toward lower values across all archetypes, with the mean decreasing by approximately 30% and the interquartile range (IQR) reduced by 29.8% compared to the baseline scenario, indicating that adopting off-site prefabrication and modular construction can effectively reduce embodied carbon.

4.3. From Buildings.city to Buildings.sg: A scalable web platform and toolkit

4.3.1. Platform interface and companion package

Fig. 12 presents the structure of the proof of concept web platform (Buildings.sg), which integrates a navigation system for both empirical and simulated data. User-configurable toggles and filters include building archetype classification and an optional display of buildings with machine learning-predicted features. A hierarchical results panel displays energy simulation outputs, including both embodied and operational carbon across archetypes, from input parameters to scenario-based results. Additionally, the platform supports data export (e.g., geospatial datasets and IDF files) for more advanced (and customized) energy simulations, which facilitate research and decision-making for policymakers, urban planners, and building energy researchers.

As illustrated in the interface (Fig. 13), the platform offers a

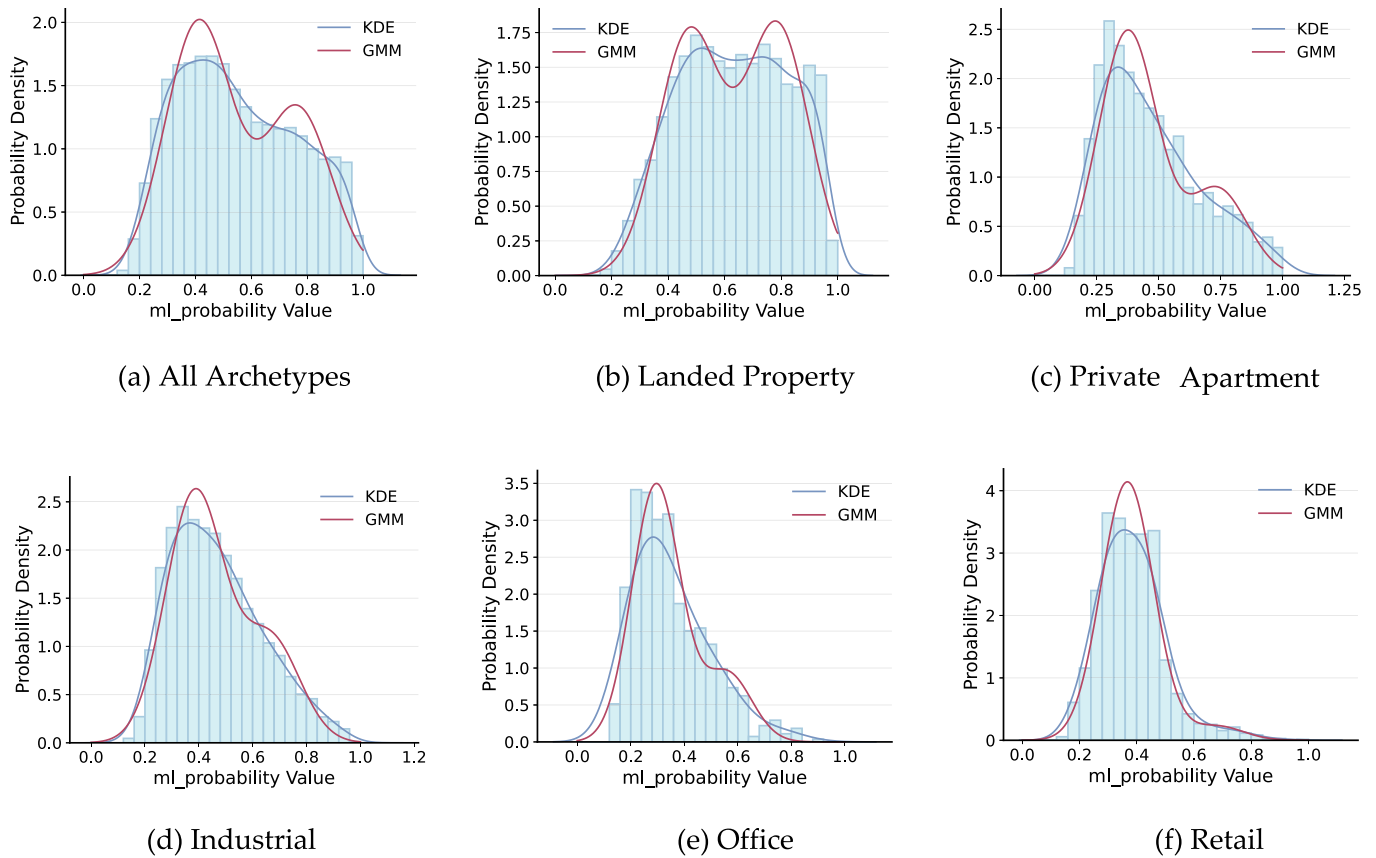


Fig. 9. Probability density distributions of predicted probabilities by building archetype.

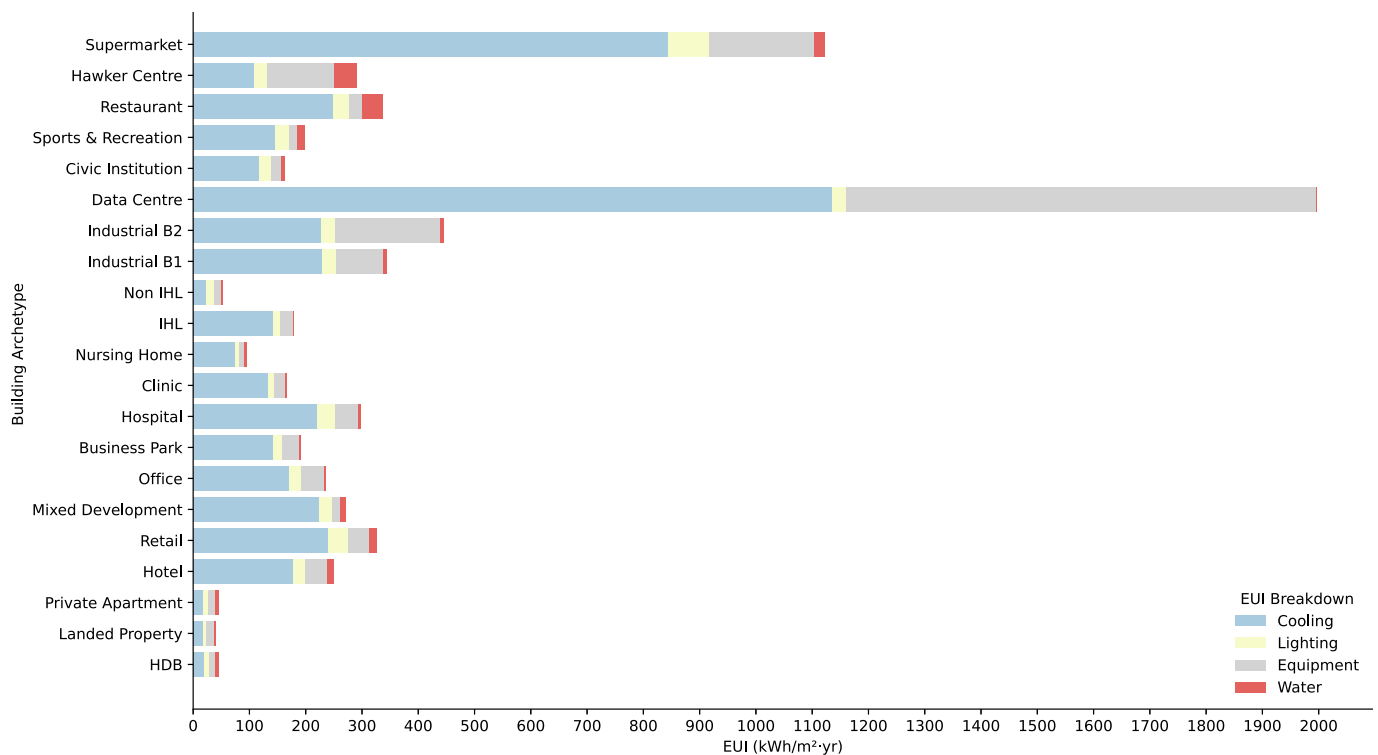


Fig. 10. Overview of baseline energy simulation results for various archetypes.

streamlined yet comprehensive user experience. Built on a Mapbox base map, it features collapsible side panels (viz., summary charts and filters

on the left enable scenario and archetype comparisons, while the right panel provides data downloads and additional context). Users can click

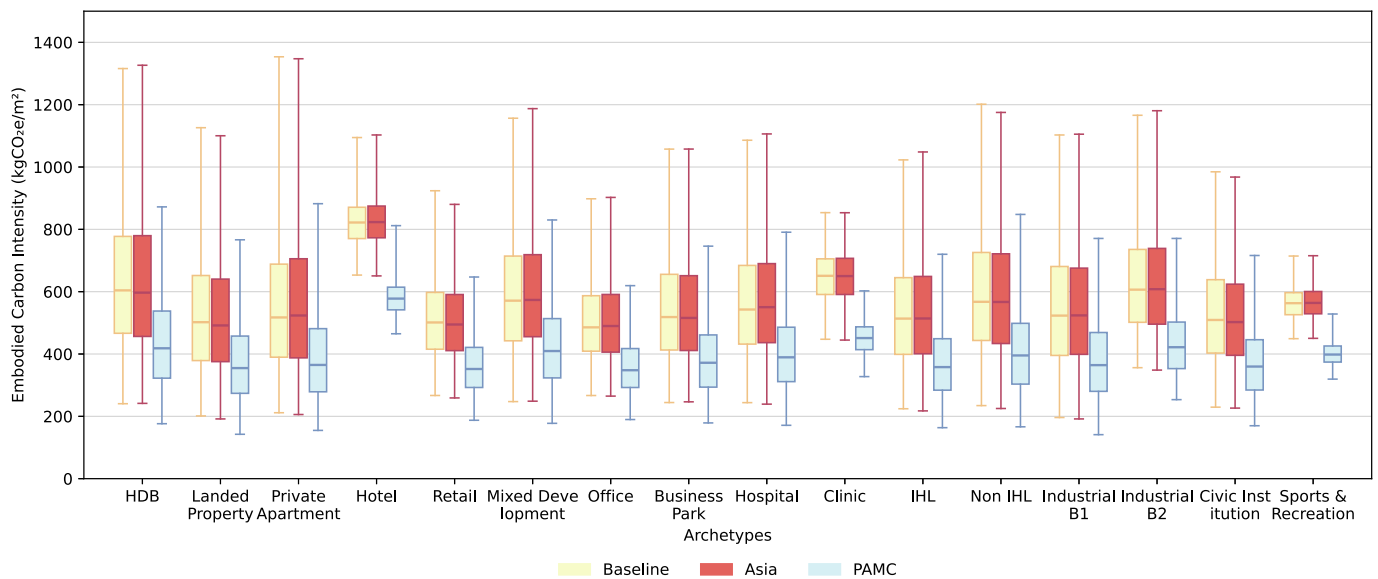


Fig. 11. Comparison of embodied carbon intensity across archetypes from Monte Carlo simulations (based on data trimmed at both the top and bottom 2% of 5000 samples).

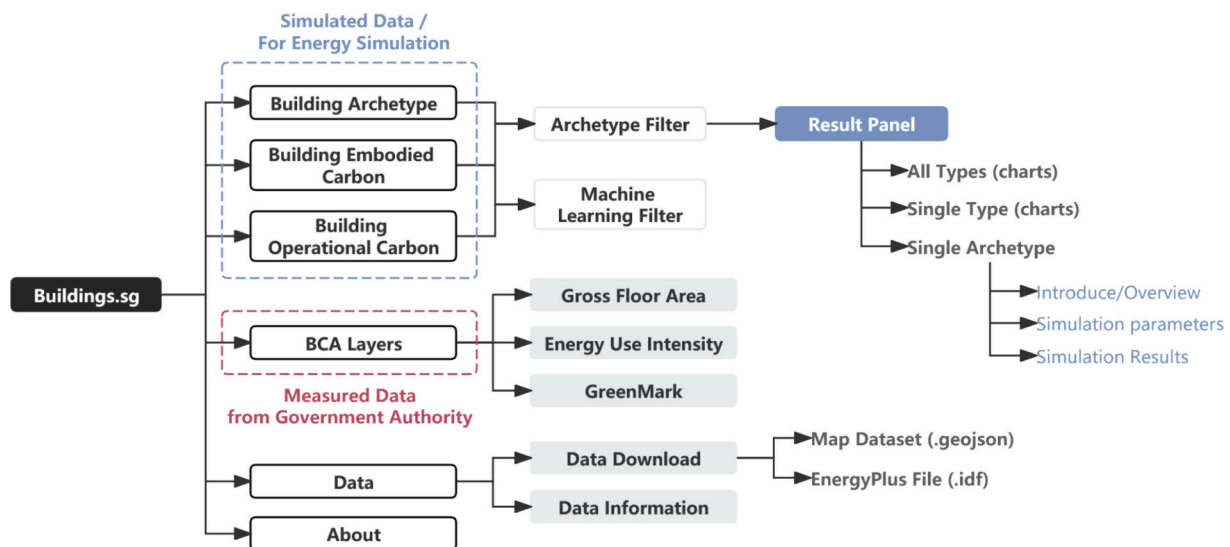


Fig. 12. The Buildings.sg (and Buildings.city) platform structure/framework.

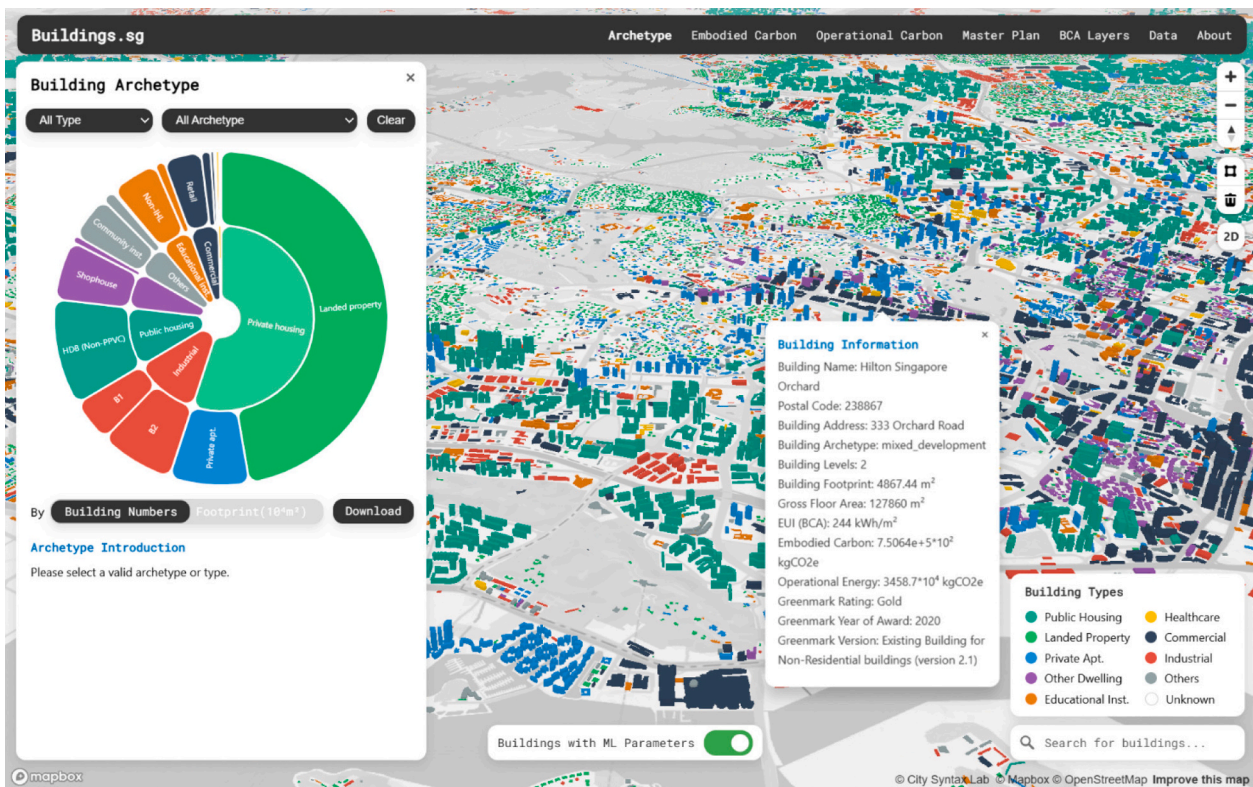
individual buildings to access simulation results, measured data, archetype predictions, and related attributes.

To enable municipalities worldwide to rapidly deploy their own interactive UBEM environments, the core of this research is built upon Buildings.city, a lightweight, Vite-based generalized package. While Buildings.sg serves as a comprehensive local proof of concept, the underlying Buildings.city framework is optimized for rapid development and efficient modular handling. By externalizing the configuration (config.json) and dataset, the toolkit supports a data-driven workflow in which city-specific building data (e.g., archetypes, operational energy, and embodied carbon) can be integrated with minimal modification. Notably, the toolkit/package allows users to run the back-end Random Forest archetype classification model to predict unknown archetype values for their own datasets. The full codebase is hosted on a public GitHub repository (<https://github.com/City-Syntax/buildings.city>), facilitating community-driven adaptation to other urban contexts.

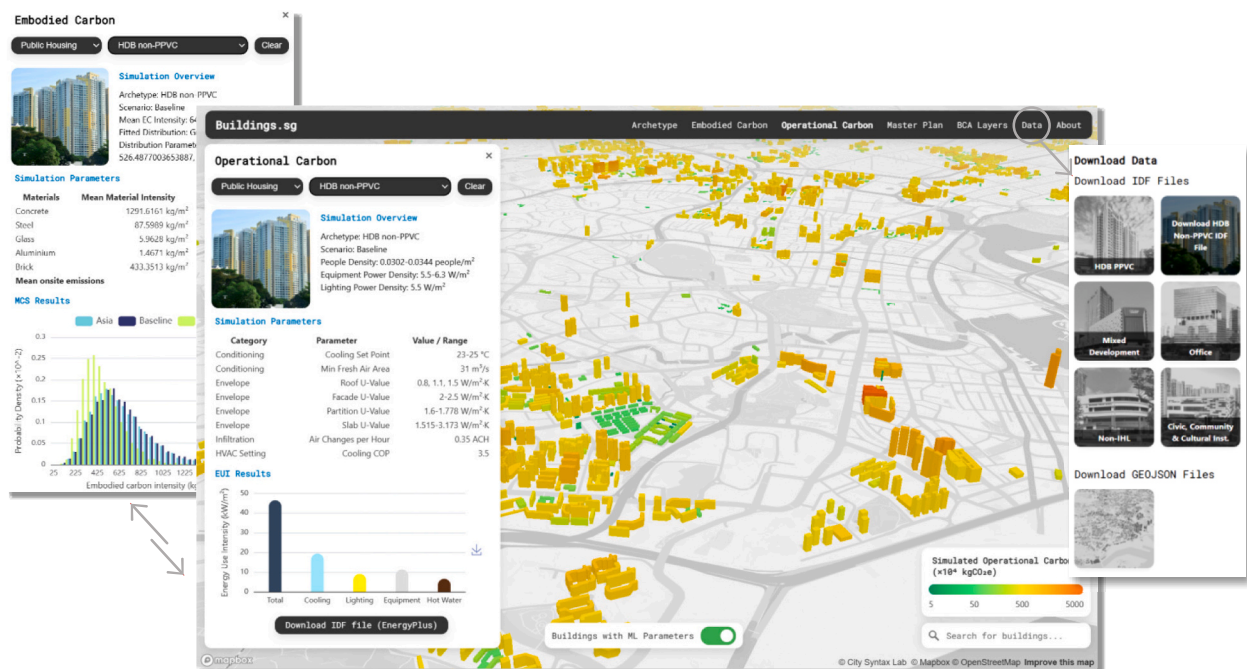
4.3.2. Scalable simulation pipeline

Operational energy simulation is implemented via a tightly integrated Grasshopper-Python-EnergyPlus workflow. Grasshopper clusters perform geometric processing, parameter mapping, and generation of archetype-specific OpenStudio job structures. Each archetype cluster encapsulates schedules, internal loads, HVAC parameters, and weather bindings, ensuring consistent simulation logic across the region. The workflow inputs include the Rhino3D model of the region with archetype assignments and the relevant weather files, while outputs consist of building-level EUIs and end-use intensity breakdowns. Python scripts coordinate batch EnergyPlus execution, parse outputs, and assemble regional energy results. All clusters and scripts adopt a modular architecture with version-controlled definitions, enabling full reproducibility across regions. Fig. 14 illustrates the complete UBEM energy-simulation pipeline.

In parallel, the embodied-carbon estimation workflow enriches the unified building data model with construction-system attributes (viz. structural system, façade type, roof assembly, material layers) drawn



(a) Building archetype layer with all buildings displayed, showing enabled structures with machine learning parameters and detailed information of a selected building.



(b) Result Panel: Operational carbon layer without filter applied and embodied carbon layer with HDB non-PPVC filtered. The Data link supports downloading both IDF and GeoJSON files.

Fig. 13. Visualizing UBEM: interface screenshots of Buildings.sg platform.

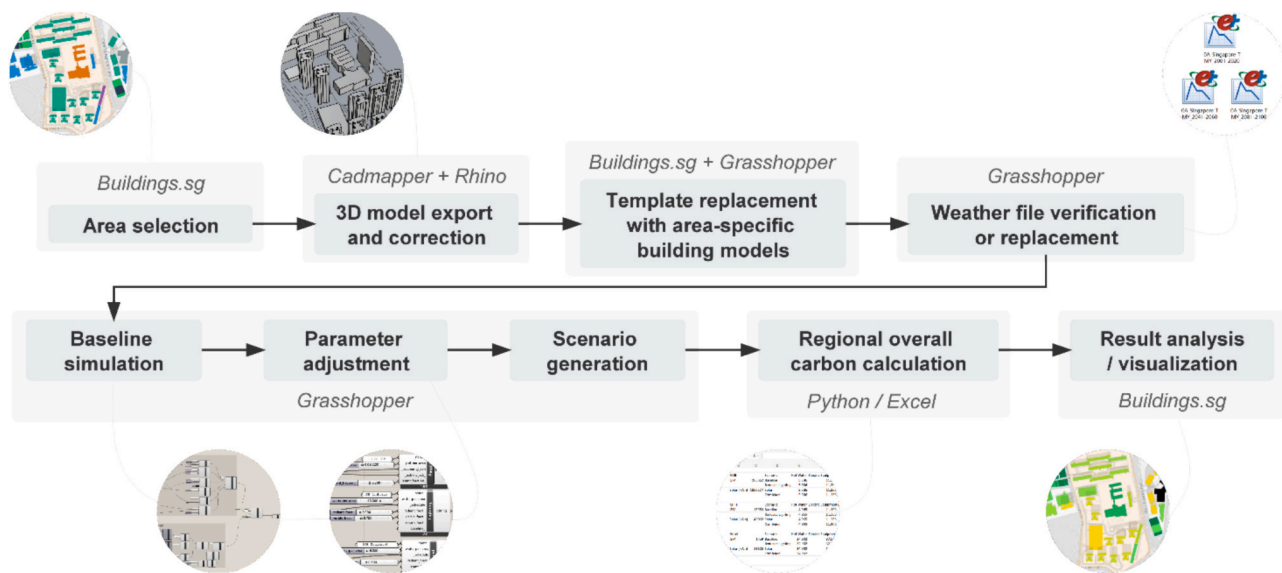


Fig. 14. UBE energy-simulation pipeline for urban districts.

from national databases or assigned via archetype defaults. A dedicated Grasshopper component converts these attributes into a Bill of Materials (BOM) representation linked to emission factors from open LCA datasets. Python scripts then aggregate quantities, compute cradle-to-gate embodied carbon, and align the results with operational-energy outputs. The workflow, with fixed schemas for inputs, outputs, and intermediate files, allows reproducible recalculation under alternative material assumptions or LCA datasets. Both pipelines can be reproduced following the technical documentation (<https://city-syntax.github.io/buildings.sg/documentation.html>), enabling scenario-based analyses for other urban districts.

5. Discussion

5.1. Practical implications

UBEM research has traditionally focused on isolated components – such as data acquisition (Davila et al., 2016) or simulation workflows (Zhang et al., 2025) – without a cohesive framework that targets downstream application, policy usability, and impact. This study addresses fragmentation by introducing a unified, open-source platform and framework that spans data preprocessing, machine-learning-based archetype classification, energy simulation, and interactive geospatial visualization. Our platform embodies this integration, providing an end-to-end pipeline that supports both academic research and urban policy planning.

5.1.1. Geospatial transferability and ML-enhanced archetype assignment

The (automated) prediction of building archetypes based on geometric and spatial features alleviates the dependence on manual classification, which often hampers scalability in UBE projects (Ang et al., 2020). Moreover, because the model relies on transferable geometric features rather than city-specific building identifiers, it can be readily applied to other urban areas with sufficient OSM coverage, facilitating cross-city comparisons and broader scalability.

5.1.2. Policy-relevant carbon mapping and reproducibility

The platform's ability to visualize both operational and embodied carbon emissions at the individual-building level enables policymakers to identify high-emission zones and prioritize interventions, such as retrofits or energy-efficiency upgrades. By making the entire pipeline open-source – including EnergyPlus-ready IDF templates, training

scripts, and annotated geospatial datasets – this study addresses the reproducibility gap that has limited the policy uptake of previous UBE research. Practitioners can now replicate or extend the framework with transparent assumptions and parameter definitions (rather easily and quickly).

5.1.3. Modular design for contextual adaptations

While the current comprehensive proof of concept uses Singapore as the study region, the system's modular design allows users to plug in alternative inputs, adapting the framework to other cities or climate zones. To explicitly demonstrate this geographical transferability, the Buildings.city repository includes a baseline deployment example for Zurich. By simply externalizing the configuration dataset and substituting tropical weather files and local parameters with European archetype templates and Zurich's OpenStreetMap data, the platform can map a completely disparate urban environment. Local codes, datasets, and energy benchmarks can be substituted without modifying the underlying codebase, offering flexibility for regional calibration and future scenario testing.

5.1.4. Enhancing stakeholder engagement and democratic data access

Our framework (platform) serves as a "common digital language" for diverse stakeholders. In a fragmented governance landscape, a unified web-based interface democratizes access to carbon emissions and performance data, and we also provide a full set of UBE simulation templates that industry (e.g., consultants) and policymakers can use for baseline testing/benchmarking and comparison. This fosters inter-agency and/or industry collaboration and enables non-expert stakeholders (such as building owners or community groups) to engage with complex energy data through intuitive, spatially contextualized 3D interfaces. Such accessibility is essential for fostering the public-private partnerships required to meet ambitious climate and carbon emissions reduction goals.

5.2. Limitations and future work

Despite the advances presented in this study, several limitations remain. First, the use of open-source OSM data introduced missing values in height and archetype attributes, particularly for (underrepresented) categories such as clinics and restaurants, affecting classification performance. While the Random Forest model improved label coverage, the limited training dataset constrained its accuracy. Second,

simulations relied on static climatic inputs and generalized occupancy schedules, overlooking behavioural diversity and temporal variation, which can substantially affect energy demand. Third, the current platform lacks real-time and demand-response modeling capabilities, limiting dynamic analysis. Future work can prioritize (further) collaboration with local authorities for enhanced data access, integrate agent-based modeling to capture occupant behaviors, and implement cloud-based parallel simulation tools and digital twin integration to support adaptive, city-scale planning. These improvements will further enhance the framework's utility for evidence-based urban decarbonization strategies.

6. Conclusion

This study establishes Buildings.city, an open-source framework (and platform) that addresses critical gaps in urban-scale building energy modeling and carbon emissions assessment for global municipalities. To validate the framework, we deployed a city-scale proof of concept in Singapore (Buildings.sg), where we resolved archetype incompleteness using supervised machine learning with over 75% accuracy, combined with multi-source geospatial integration. Templates were developed for 23 building archetypes and applied to district- and city-level 3D models using Grasshopper and Python. This system enables city-wide, spatial evaluation of both operational and embodied carbon across approximately 120,000 buildings.

The platform's interoperability, through standardized inputs and outputs (EnergyPlus-ready IDF, GeoJSON datasets) and a modular design, advances UBEF reproducibility beyond that of proprietary tools. Beyond its technical contributions, the framework and platform's transparency and scalability align with global calls for collaborative urban climate action (UNEP, 2023), offering policymakers a robust baseline tool for benchmarking, scenario testing, and emissions tracking.

Future work can incorporate dynamic occupancy models and enhanced backend computational capabilities to improve accuracy and scalability. Deeper collaboration with government agencies will be essential to access high-resolution datasets that can support more precise calibration and wider applicability. Ultimately, the framework and platform can inform decarbonization pathways and guide evidence-based energy policy at both the district and national levels.

Appendix A. Demonstration Case Study using Buildings.sg: district-level building energy modeling in Singapore

As illustrated in the flowchart (Fig. 14), the demonstration case study process begins with defining a study area (of interest) using the polygon selection tool on the Buildings.sg platform. Upon selection, users receive summary statistics – including the number of buildings by archetype, average EUI, and total carbon emissions – which inform the scope and characteristics of the study area. The selected district can be exported in Rhino3D-compatible format (e.g., via standard online tools like Cadmapper), with minor adjustments made as needed. Missing attributes, such as storey count or height, can be cross-referenced directly from individual buildings on the platform.

Corresponding archetype IDF templates can then be downloaded (from Buildings.sg) and integrated into Grasshopper to replace the default models. After verifying the meteorological inputs, baseline energy simulations are conducted. Scenario variations are created by adjusting key parameters, such as building envelope properties and occupancy schedules. Results are reported by archetype, detailing total EUI and its end-use components (e.g., cooling, lighting, equipment, and hot water). The overall energy consumption for the district is computed by aggregating archetype-level results using the following formula:

$$\text{Regional EUI} = \frac{\sum_{i=1}^k \text{EUI}_i \bullet A_i}{\sum_{i=1}^k A_i} \quad (8)$$

Where k is the total number of building types in the region; A_i is the Gross Floor Area (GFA) of building type i ; and EUI_i is the Energy Use Intensity (kWh/m²·year) for building type i , corresponding to a specific energy end-use. This formulation can be independently applied to different end-use categories, such as cooling, lighting, equipment, and hot water. Photovoltaic (PV) savings, expressed as negative EUI values, may be incorporated in the same manner to estimate net energy use or assess net-zero energy performance.

Singapore's tropical climate imposes significant cooling demands on the built environment, particularly in high-density public housing and

Code and data

The generalized UBEF framework and toolkit Buildings.city, is available at <https://Buildings.city> or <https://github.com/City-Syntax/buildings.city> (current release v1.0). This repository enables the rapid deployment of comparable urban carbon mapping platforms across diverse global contexts using configurable GeoJSON datasets and parameters, and includes a baseline deployment example for Zurich. The comprehensive city-scale proof of concept platform for Singapore is publicly accessible at <https://buildings.sg>, with its specific source code and documentation available on GitHub (<https://github.com/City-Syntax/buildings.sg>). The platform includes EnergyPlus-compatible IDF templates for simplified and detailed models, and GeoJSON datasets for GIS-based analysis. The current release (v1.0) incorporates building dataset v5.0, energy simulation v5.0, and embodied carbon simulation v3.0. Both repositories are distributed under the MIT License, permitting free use, modification, and distribution.

CRedit authorship contribution statement

Pengdi Lyu: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Hongyao Wong:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis. **Royden Soh:** Writing – review & editing, Software, Formal analysis. **Tao Wang:** Writing – review & editing. **Yu Qian Ang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

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institutional buildings. In support of national decarbonization goals, initiatives such as the Green Towns Programme and public sector energy efficiency efforts have encouraged rooftop PV and system-level retrofits. This demonstration case study focuses on the Chinatown Cantonment Rd/Neil Rd area (Fig. A1), comprising approximately 22 buildings across four archetypes, including a primary school (non-IHL building archetype), two HDB clusters to the north and south, private apartments, and a hotel.

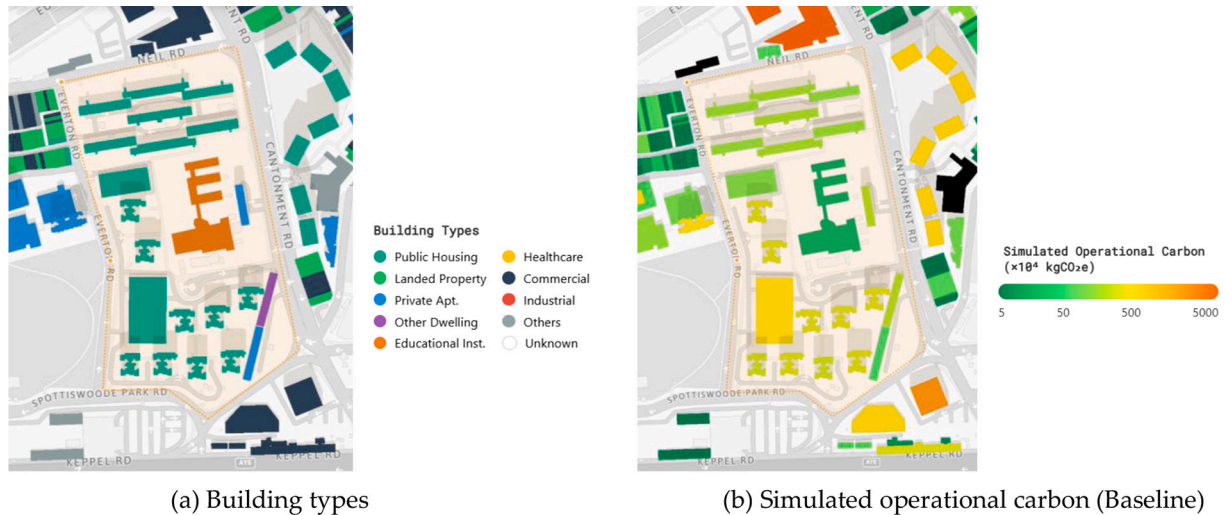
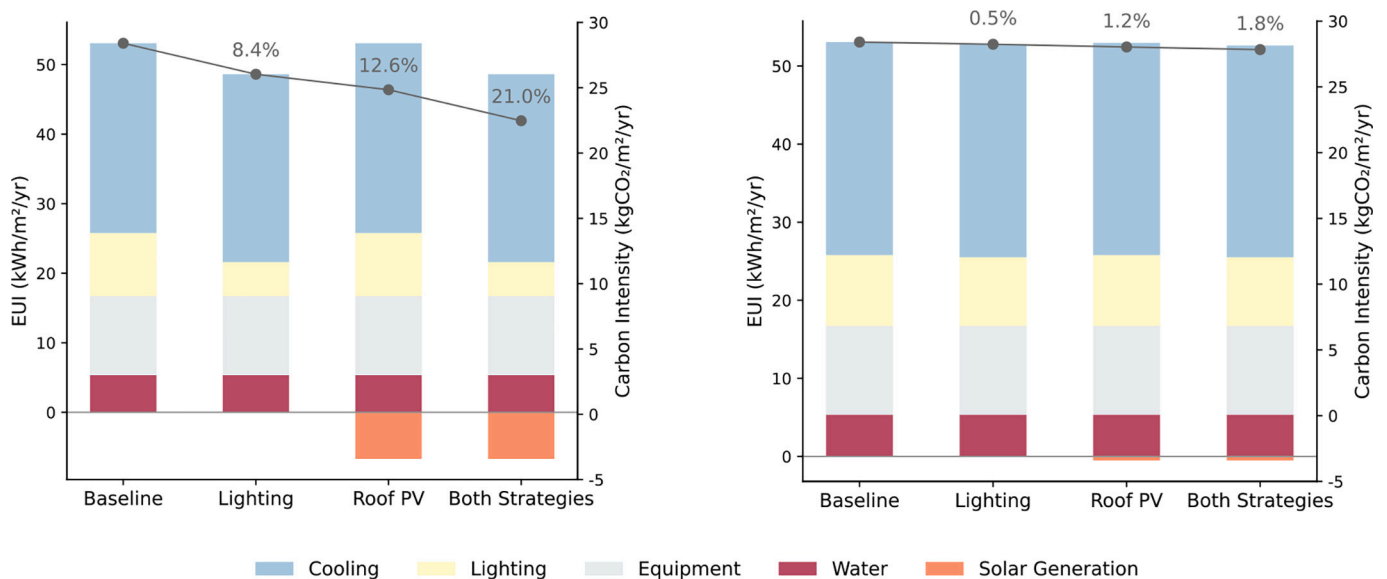


Fig. A1. Demonstration case study of the Chinatown, Cantonment Rd/Neil Rd area (orange polygon area) (screenshots from Buildings.sg).

With the baseline model completed and validated, three scenario assessments were developed. The first scenario explores a simpler, cost-effective strategy aimed at complying with the lighting requirements of the BCA Green Mark Platinum standard (the highest tier in Singapore's green building certification scheme). The second scenario examines the impact of rooftop PV systems, in line with recent policy initiatives promoting their adoption in HDB. To simulate these scenarios, two key parameters were adjusted across various building archetypes: lighting power density (LPD) and on-site electricity generation from rooftop PV. In the residential sector, both HDB flats and private apartments were initially modeled with identical baseline assumptions. The LPD was reduced from 10.0 W/m² to 5.0 W/m², and rooftop PV was assumed to cover 40% of the total roof area, with electricity generation simulated using hourly solar irradiance data over a full year. The third scenario combines both strategies to evaluate their cumulative impact.

The impact of different energy-saving strategies across the two building archetypes on overall district EUI and operational carbon intensity is illustrated in Fig. A2. Results show that interventions targeting the HDB archetype led to more significant district-wide carbon reductions, due to their large share of total floor area. In contrast, the impact on private apartments is minimal, with the combined strategy reducing carbon intensity by only 1.8%. Among the single measures, HDBs' rooftop PV is most effective, achieving a 12.6% reduction in operational carbon emissions. The combined strategy enhances this effect, reducing emissions by 21.0%. Applying efficient strategies to dominant archetypes significantly improves district performance, even if some individual buildings consume more energy. These findings support the Green Towns Programme by highlighting the effectiveness of archetype-specific retrofits in achieving scalable urban decarbonization.



(a) Strategies for HDB

(b) Strategies for private apartment

Fig. A2. Comparative effects of energy-saving strategies on district-level EUI and operational carbon.

Data availability

Data used is open source on the platform and github repo

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